



ULUSLARARASI KIBRIS ÜNİVERSİTESİ
CYPRUS INTERNATIONAL UNIVERSITY

Project Title: Parking Slot Reservation System

CMPE314 – SOFTWARE ENGINEERING

PROJECT PROPOSAL

Submitted To: Asst. Prof. Dr. Felix Babalola

Group Number:

Submitted By:

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1. INTRODUCTION

1.1. Project Background

Parking has become a major challenge in universities, shopping centers, offices, hospitals, and residential areas due to the increasing number of vehicles. Drivers often search manually for available parking spaces, wasting time and causing congestion. A digital reservation system can improve efficiency by allowing users to reserve spaces in advance.

1.2. Problem Statement

There is no centralized system that allows drivers to check parking availability and reserve slots beforehand. This results in wasted time, disorganization, and inefficient parking management.

1.3. Project Aim

To develop a web-based Parking Slot Reservation System that enables users to view, reserve, and manage parking slots efficiently.

2. PROPOSED SOLUTION

2.1. Description of the System

A web-based system where users can register, log in, view available parking slots, reserve them, and receive confirmations. Admins can manage slots and monitor usage.

2.2. Key Features (High-Level)

- Feature 1: User Registration and Authentication
- Feature 2: View Available Slots
- Feature 3: Online Reservation
- Feature 4: Modify/Cancel Booking
- Feature 5: Admin Dashboard
- Feature 6: Notifications

- Feature 7: Reports

3. STAKEHOLDERS AND TARGET USERS

3.1. Primary Users

- **Drivers:** Reserve parking easily.
- **Parking Admins:** Manage slots.

3.2. Secondary Stakeholders

- **System Administrator:** Manages user permissions and system backups.
- **Facility Management:** Needs to generate reports on parking slots usage.
- **Security Staff**

4. REQUIREMENT HIGHLIGHTS

ID	Requirement Description	Priority (High/Med/Low)
FR-01	The system shall allow users to create a new account.	High
FR-02	The system shall allow users to Login by title.	High
FR-03	The system shall display the availability status of slot.	High
FR-04	The system shall allow the user to reserve slots.	High
FR-05	The system shall user to Cancel/Modify slots.	Medium
FR-06	The system shall send a due-notifications reminder.	Medium

FR-07	The system shall allow the admin to add new slots, release slots, fix status on maintenance and make available slots records.	High
FR-08	The system shall keep reports of slots used and with user details.	Medium

5. TECHNOLOGY STACK

Layer	Technology Choice	Justification
Frontend (UI)	HTML, CSS, JS	Provides a dynamic user experience with the interface.
Backend (Logic)	Django/Node.js	Team has prior experience
Database	MySQL/PostgreSQL	Suitable for structured database management
Version Control	Git	For collaboration and code backup.
Open Source Libraries	Bootstrap	To speed up development.

6. PROJECT PLAN AND TIMELINE

6.1. Task Breakdown & Gantt Chart

Phase	Week	Activities	Deliverable
Phase 1	Week 1-2	Requirements Elicitation (interviews, surveys)	SRS Draft
Phase 2	Week 3-4	UML Modeling & Design	Complete UML Diagrams
Phase 3	Week 5-6	Implementation (Core Features)	Working Prototype
Phase 4	Week 7-8	Testing, Documentation, & Final Touches	Final Report & Presentation



6.2. Team Responsibilities

Team Member	Primary Responsibility
Samuel Akpoghene Otobo	Requirements & Use Case modeling
22100896	Class Diagram & Database Design
	UI Design & Frontend Development
	Backend Logic & Testing
	Documentation & Project Coordination

7. CONCLUSION

This system improves parking efficiency and demonstrates practical software engineering skills.